Paper F-03: Black Hole Thermodynamics: A Constraint-Driven Flux Dynamics (CDFD) Perspective

Steve Bico Mujjabi, MD
Independent Researcher
Founder, Vura Labs
Kampala, Uganda
ORCID: 0009-0001-0556-5516

May 2026

Abstract

This paper develops a falsifiable CDFD/AFL model of Black Hole Thermodynamics. It maps mass, energy, angular momentum, magnetic flux, and accretion to driving flux Φ, horizon geometry, gravity, radiative efficiency, and disk transport to active constraint C, disk reconfiguration, jets, outflows, and environmental feedback to responsiveness S, and spin, charge, growth, magnetic topology, and accretion history to structural memory M_s. The target outcome is luminosity, growth, jet power, and thermodynamic accounting, tested against gravitational waves, spectra, timing, imaging, accretion estimates, and simulations. The paper treats universality as a shared model architecture rather than an assertion that unlike systems have the same material mechanism. It states the negative result that would make the mapping unnecessary and places the domain claim inside the cumulative CDFD programme and its reproducible runtime boundary.

Symbol Conventions

CDFD/AFL Part IV symbol convention.

Symbol	Part IV meaning	Cross-domain reading
Φ	Driving flux or throughput	Energy, matter, signal, capital, information, traffic, force, or biological load entering a system
C	Active constraint or capacity burden	Bottleneck, resistance, boundary, scarcity, toxicity, regulation, topology, latency, or load-bearing limit
S	Responsiveness or adaptive surface	The system's ability to reconfigure, buffer, route, repair, learn, or preserve function under changing load
M_s	Structural memory	Stored damage, hysteresis, inherited topology, institutional memory, trained weights, ecological legacy, or retained state
Ψ_s	Adaptive operating ratio $(\Phi/C) \cdot S \cdot M_s$	Local bookkeeping gauge for constrained operation, overload, recovery, or regime change; not a universal constant
Λ	Life Number or capstone surplus ratio where explicitly defined	Reserved for Part II-derived life-number notation or synthesis-level surplus measures, never an undefined decoration

Greek-symbol convention.

Symbol	Local role	Interpretation rule
α	Accumulation or reinforcement coefficient	Sets how fast a pathway, constraint, habit, cascade, or retained structure grows under load
β	Relaxation, decay, clearance, or washout coefficient	Sets loss, recovery, damping, forgetting, degradation, or return toward baseline
γ	Coupling, diffusion, redistribution, or transfer coefficient	Sets spread across nodes, layers, tissues, markets, infrastructures, media, or physical phases
δ, ϵ	Perturbation, tolerance, small offset, or numerical guard	Used only as local thresholds, stability margins, measurement tolerances, or stress increments
η, λ, μ, ν	Efficiency, rate, baseline, intensity, or frequency terms	Meaning is fixed by the equation, subscript, and domain-specific proxy in the paragraph where the term appears
ρ, σ, τ, ϕ	Density/correlation, variability/noise, time-scale, phase, or field terms	Local coefficients only; subscripts bind them to the named pathway, domain, or experiment
Ω, Λ	Domain/state space; Life Number where defined	Ω names a modeled state space; Λ carries life-number or synthesis notation only when the paper defines it

1 Series Context

Part I sets the flow–constraint language used here [2]. Part II develops the Life Number and the origins-of-life use of the same notation [3]. Part III applies AFL to biology and medicine

[4]. Part IV [5], DOI `https://doi.org/10.5281/zenodo.20395901`, is the cross-domain stress test: it asks where the notation clarifies measurement, and where it becomes only a metaphor.

2 Introduction

2.1 The Mujjabi Universal Capacity Law

Building directly upon the Fundamental Physics (Part I) [2], the **Mujjabi Universal Capacity Law** proposes that across all scales—from black hole event horizons to global financial markets and power grids—driving high flux (Φ) against a rigid topological constraint (C) can drive system saturation. When Φ/C approaches critical thresholds, the system does not scale linearly; it undergoes a topological phase transition.

2.2 The Mujjabi Universal Memory Law

The **Mujjabi Universal Memory Law** frames persistent systemic collapse. When a complex network fails to clear its structural memory (M_s) and its relaxation parameter decays ($\tau_{relax} \to \infty$), temporary stressors become permanent rigidities. This is the proposed CDFD mechanism behind ecological death, economic depressions, and infrastructural gridlock.

In standard relativistic frameworks, a black hole is a region of spacetime exhibiting such strong gravitational acceleration that nothing, including light, can escape. The boundary defining this region is the event horizon. Following the works of Bekenstein and Hawking, the event horizon is understood to possess an entropy proportional to its surface area, $S_{BH} = \frac{k_B A}{4l_p^2}$, suggesting that all the information of the infalling volume is holographically encoded on the 2D boundary.

The CDFD paradigm posits that the universe is governed by the continuous interaction of flow (Φ) and constraint (C) across adaptive surfaces. In this context, a black hole is not a singularity of broken physics; rather, it is the maximal boundary condition of the CDFD equations. It is the extreme limit where constraint C approaches infinity for classical flow, forcing all throughput to undergo quantum phase transitions.

3 The Event Horizon as an Active Surface

In the CDFD framework, any boundary separating two topological regions evaluates to an active surface governed by the equation:

$$\Psi_s = \left(\frac{\Phi}{C}\right) \cdot S \cdot M_s \tag{1}$$

For an event horizon, the parameters map as follows:

- **Flux (Φ)**: The accretion of matter and information crossing the boundary (inward), and Hawking radiation (outward).
- **Constraint (C)**: The causal barrier generated by extreme spacetime curvature. For macroscopic inward flow, the boundary allows absolute transmission. For outward classical

flow, $C \to \infty$.

- **Responsiveness (S)**: The geometric surface area of the horizon A, which defines the thermodynamic capacity to process and encode incoming flux.
- **Memory (M_s)**: The holographic encoding. Every bit of information that crosses the horizon is chronicled on the surface. Therefore, the event horizon possesses perfect, unbroken memory, $M_s \to 1.0$.

3.1 Information Bottlenecks and The Bekenstein Bound

As matter falls into the black hole, the inward flux Φ_{in} causes the mass, and thus the surface area S, to increase. This constitutes a dynamic adaptation. The horizon expands its surface responsiveness precisely to accommodate the increased memory load M_s. The Bekenstein bound is essentially the limit of information density allowed by the constraint C.

If the black hole attempts to absorb flux faster than the surface can expand and encode it, the system would theoretically experience a $\Psi_s > 1$ "over-flux" anomaly. However, the geometric rigidity of General Relativity ensures that S expands in lockstep with mass, maintaining $\Psi_s \approx 1$.

4 Hawking Radiation as Regime Stabilization

If an isolated black hole ceases to accrete matter ($\Phi_{in} = 0$), the static nature of the system threatens to drop the adaptive ratio $\Psi_s \ll 1$. In biological systems (AFL), a drop in throughput leads to decay. In the cosmological vacuum, the event horizon undergoes analogous decay to shed its memory load and lower its constraint profile.

Hawking radiation (Φ_{out}) emerges as the necessary quantum leakage required to satisfy the CDFD equation. To prevent $\Psi_s \to 0$, the vacuum surface generates an outward flux at the expense of its own structural memory (mass/area). As the horizon shrinks, the temperature increases, amplifying Φ_{out} until complete evaporation occurs.

5 Measurement Layer

The first job in any domain is to choose proxies before drawing conclusions. Φ may be a rate of material, energy, information, capital, or signal flow. C is the bottleneck that slows or redirects that flow. S measures the system's ability to reconfigure under stress, and M_s records the past load that still changes present behavior. A useful application gives those four terms observable definitions, a time scale, and a failure condition. If a standard domain model explains the same data more cleanly, the CDFD/AFL mapping should give way.

5.1 Current Runtime Diagnostic: Universal Network Cascade

The release-local discovery script keeps this paper tied to the current CDFD Runtime [6], including RK4 field updates, surface dynamics, structural-memory diffusion, and a finite-value audit. In the 2026-06-04 runtime pass, the localized hub-drive stress test ended with hub

$\Psi_s = 5.21$, peak network $\Psi_s = 5.21$, 21 nodes above $\Psi_s > 1.5$, and 37 maximum memory-locked nodes. I treat those numbers as a runtime sanity check and as a target for later domain measurement, not as a substitute for field data.

6 Major Universal Upgrade: Mechanism, Scale, and Tests

6.1 Observable Translation

The paper-specific translation begins with an observable chain rather than a verbal analogy. For Black Hole Thermodynamics, the proposed drive is mass, energy, angular momentum, magnetic flux, and accretion. The active limitation is horizon geometry, gravity, radiative efficiency, and disk transport. Responsiveness is carried by disk reconfiguration, jets, outflows, and environmental feedback, while retained state is represented by spin, charge, growth, magnetic topology, and accretion history. The outcome to explain is luminosity, growth, jet power, and thermodynamic accounting.

Table 1: Operational CDFD/AFL map for Paper F-03.

Term	**Paper-specific observable meaning**	**Measurement question**
Φ	mass, energy, angular momentum, magnetic flux, and accretion	What rate, load, gradient, or demand enters the focal system?
C	horizon geometry, gravity, radiative efficiency, and disk transport	Which measured bottleneck limits transfer, function, or recovery?
S	disk reconfiguration, jets, outflows, and environmental feedback	Which process changes effective capacity or reroutes the load?
M_s	spin, charge, growth, magnetic topology, and accretion history	Which retained state makes equal present loads produce different futures?
Y	luminosity, growth, jet power, and thermodynamic accounting	Which outcome is predicted out of sample, with uncertainty?

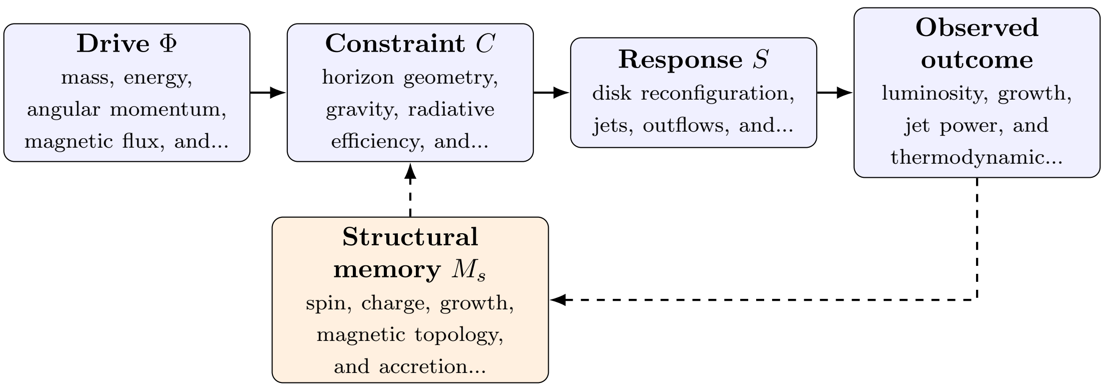

Figure 1: Paper F-03 observable CDFD/AFL architecture for Black Hole Thermodynamics. Solid arrows give the proposed forward chain; dashed arrows make history-dependent feedback explicit.

6.2 Minimal Coupled Model

A paper-level model must separate throughput, constraint accumulation, response, and history. One deliberately minimal form is

$$Y(t) = \frac{\Phi(t)S(t)}{\epsilon + C(t)}, \tag{2}$$

$$\frac{dC}{dt} = \alpha\Phi(t) - \beta S(t)C(t) + \gamma\mathcal{L}C(t), \tag{3}$$

$$\frac{dM_s}{dt} = \eta C(t) - \mu M_s(t), \tag{4}$$

$$\Psi_s(t) = \frac{\widehat{\Phi}(t)}{\epsilon + \widehat{C}(t)}\,\widehat{S}(t)\,[1 + \omega\widehat{M_s}(t)]. \tag{5}$$

Here Y is the measured output, $\epsilon > 0$ prevents a singular denominator, $\mathcal{L}$ is a spatial or network coupling operator, and hats denote explicitly normalized variables. The sign and magnitude of ω must be estimated: memory can preserve useful organization or amplify damage. No fixed threshold is universal before the variables, normalization, and observation scale are declared.

For this paper, the hypothesized causal sequence is specific. A change in mass, energy, angular momentum, magnetic flux, and accretion arrives faster than disk reconfiguration, jets, outflows, and environmental feedback can compensate. This raises or redistributes horizon geometry, gravity, radiative efficiency, and disk transport. If the load persists, the retained state encoded by spin, charge, growth, magnetic topology, and accretion history changes the next response, so a later event of equal magnitude can produce a different trajectory in luminosity, growth, jet power, and thermodynamic accounting. The scientific task is to estimate that sequence against the best ordinary model of the domain, not merely to redescribe the outcome after it occurs.

6.3 Cross-Scale Universality Without Mechanistic Erasure

For the cosmic and subatomic systems family, the universal comparison must remain subordinate to conservation laws, relativity, quantum mechanics, and established astrophysical or plasma equations. CDFD is useful only if it compresses a regime transition or hysteresis question without pretending that biological adaptation and physical response are the same mechanism. [8, 7, 1] The CDFD programme is cumulative: Part I supplies the flow–constraint foundation [2]; Part II asks when maintained throughput supports persistent organized states [3]; Part III develops adaptive limitation and biological memory [4]; and Part IV [5] tests how much of that architecture survives translation. The CDFD Runtime [6] supplies a reproducible numerical stress surface, but it does not substitute for the domain evidence named here.

The required scale declaration for this paper is microseconds to cosmic time; horizons to galactic environments. At short scales, the key question is whether the incoming drive is transmitted, buffered, or diverted. At intermediate scales, response changes effective capacity and network exposure. At long scales, retained structure can alter the state space itself. A result is genuinely cross-scale only when the aggregation rule is stated and the same conclusion is not an artifact of averaging.

6.4 Evidence Ladder and Discriminating Tests

Table 2: Evidence ladder for Paper F-03. Each rung can fail independently.

Test	**Design**	**Discriminating result**
Proxy audit	Measure gravitational waves, spectra, timing, imaging, accretion estimates, and simulations and preregister how each record maps to Φ, C, S, M_s, and Y.	The mapping fails if proxies are non-identifiable, circular, or change meaning across cases.
Load ramp	Apply or observe an accretion-state transition at matched mass and environment while estimating the dominant constraint and response time.	A threshold claim requires a reproducible nonlinear change and uncertainty bounds, not a visually chosen breakpoint.
Recovery and memory	Match present load across units with different histories, then compare recovery paths.	The memory claim requires history-dependent divergence after current conditions are controlled.
Model comparison	Compare the baseline domain model with and without the CDFD/AFL state variables using held-out data.	The paper's stated falsifier is: CDFD introduces no distinguishable prediction beyond relativity and accretion physics.

The strongest practical design combines a prospective perturbation with a recovery window. The load-ramp phase estimates where constraint begins to rise faster than output. The recovery phase estimates β and μ, separating fast relaxation from persistent memory. A topology or coupling intervention then asks whether rerouting changes the cascade as predicted. Null results are informative because they identify domains in which the universal notation adds no measurable value.

6.5 Research Programme and Boundary Conditions

The immediate research programme has four steps. First, construct a documented dataset from gravitational waves, spectra, timing, imaging, accretion estimates, and simulations and retain raw units before normalization. Second, fit a baseline domain model and report its calibration and out-of-sample error. Third, add the smallest identifiable CDFD/AFL state extension and test whether it improves prediction of luminosity, growth, jet power, and thermodynamic accounting. Fourth, repeat the analysis across the declared scale range and publish negative cases as part of the universality audit.

Three boundaries are non-negotiable. Correlation between flow and failure does not identify a constraint mechanism. A fitted memory term does not prove that the stored state has the same material meaning in another domain. Finally, a runtime cascade is evidence about the implementation, not evidence that the real system follows it. The paper becomes stronger when it can say precisely where the translation stops.

7 Conclusion

Black-hole thermodynamics and living systems can both be described with boundary, flux, and retained-state variables, but their mechanisms are not isomorphic. This paper limits the comparison to measurable accretion, horizon, spin, and radiation quantities and asks whether the CDFD translation adds anything beyond relativity and accretion physics.

Conflict of Interest

The author declares no conflict of interest.

Data Availability

Part-specific runtime diagnostics are under each Part's `outputs/` directory.

Release-wide code: `Part_E_Synthesis/supplementary/`.

Generated summaries: `Part_E_Synthesis/outputs/`.

Figures: `Part_E_Synthesis/figures/`.

8 Reference Layer

References

[1] Per Bak, Chao Tang, and Kurt Wiesenfeld. Self-organized criticality: An explanation of the 1/f noise. *Physical Review Letters*, 59(4):381–384, 1987. DOI: https://doi.org/10.1103/PhysRevLett.59.381.

[2] Steve Bico Mujjabi. Cdfd part i: Fundamental physics - constraint-driven field theory and flux dynamics, 2026. DOI: https://doi.org/10.5281/zenodo.20250821.

[3] Steve Bico Mujjabi. Cdfd part ii: Origins of life and tri-regime bioenergetics - constraint-driven flux dynamics, 2026. DOI: https://doi.org/10.5281/zenodo.20264779.

[4] Steve Bico Mujjabi. Cdfd part iii: Adaptive flux limitation biology and medicine - constraint-driven flux dynamics, 2026. DOI: https://doi.org/10.5281/zenodo.20344695.

[5] Steve Bico Mujjabi. Cdfd part iv: Universal adaptive flux limitation - constraint-driven flux dynamics, 2026. DOI: https://doi.org/10.5281/zenodo.20395901.

[6] Steve Bico Mujjabi. Cdfd runtime: Constraint-driven flux dynamics and cdfl execution engine, 2026. DOI: https://doi.org/10.5281/zenodo.20343160.

[7] Claude E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27(3):379–423, 1948. DOI: https://doi.org/10.1002/j.1538-7305.1948.tb01338.x.

[8] A. M. Turing. The chemical basis of morphogenesis. *Philosophical Transactions of the Royal Society of London. Series B*, 237(641):37–72, 1952. DOI: https://doi.org/10.1098/rstb.1952.0012.